Robot Arm Feasibility Assessment

Comparative Analysis

Kinova Gen 3 · UFACTORY xArm 7 · Universal Robots UR3

1. Introduction & Scope

This report presents a comparative feasibility assessment of three robotic arm platforms under consideration for the E-Waste Battery Extraction Robot project. The three platforms assessed are the **Kinova Gen 3**, the **UFACTORY xArm 7**, and the **Universal Robots UR3**. The assessment evaluates each platform across dimensions of simulation compatibility, software toolchain support, laboratory availability, and suitability for the intended disassembly workflow.

The overarching simulation strategy adopted by the research team consists of two sequential phases:

1. Kinematic modelling and control logic development in **MATLAB**.
2. Physics-accurate 3D simulation in **NVIDIA Isaac Sim**, leveraging URDF and configuration files to replicate real robot behaviour in a virtual environment.

The Kinova Gen 3 and UR3 are the primary simulation targets due to the broad availability of their configuration files in both MATLAB and Isaac Sim. The xArm 7, while physically present in the lab, is designated primarily as a physical validation platform.

2. Platform Overview

Kinova Gen 3

The Kinova Gen 3 is a 6-DoF collaborative robot available at the Burwood Campus Robotics Lab. Its 6-axis configuration provides high kinematic redundancy, making it well-suited for manipulation tasks in cluttered environments. URDF models and ROS packages are publicly available, enabling straightforward import into Isaac Sim. MATLAB support, while not native, is achievable via ROS bridge.

UFACTORY xArm 7

The UFACTORY xArm 7 is also a 7-DoF platform currently located in the Burwood Campus lab. It offers excellent force-torque sensing integration via the companion 6-axis sensor documented in the BoM report. Although URDF files are available, native MATLAB Robotics Toolbox integration requires additional configuration steps. The xArm 7 is designated as the physical validation and deployment platform for this project.

Universal Robots UR3

The UR3 is a compact 6-DoF collaborative robot that, while **not physically present** at the Burwood Campus, offers the most comprehensive simulation support of the three platforms. Native Robotics Toolbox models in MATLAB and ready-to-use configuration files in Isaac Sim make it the most accessible platform for early-stage kinematic development and simulation prototyping.

3. Technical Specification Comparison

Specification	Kinova Gen 3	UFACTORY xArm 7	Universal Robots UR3
Degrees of Freedom	6 DoF	7 DoF	6 DoF

Specification	Kinova Gen 3	UFACTORY xArm 7	Universal Robots UR3
Payload Capacity	4 kg	3.5 kg	3 kg
Reach	902 mm	700 mm	500 mm
Repeatability	±0.1 mm	±0.1 mm	±0.1 mm
Weight	8.2 kg	13.0 kg	11.2 kg
Communication	ROS / Ethernet	ROS / Ethernet	ROS / Modbus / EtherNet/IP
Available in Lab	Yes	Yes	No
MATLAB Support	Partial	Limited	Native (Robotics Toolbox)
Isaac Sim Support	URDF available	URDF available	Native config files
Simulation Priority	Primary	Reference	Primary (simulation)

5. End-Effector Capability Comparison

End-effector selection is critical for e-waste disassembly, where components range from large chassis panels to fine connectors and delicate PCB assemblies. The table below compares end-effector capabilities across all three platforms.

End-Effector Feature	xArm 7	Kinova Gen3 Lite	UR3
Parallel Gripper	Opening range: 86 mm	Opening range: ~100 mm	Compatible (3rd party)
Payload (gripper)	3 kg	0.5 kg	3 kg
Vacuum Gripper	78% vacuum level	Not compatible	Compatible (3rd party)
Force/Torque Sensing	6-Axis FT sensor (optional)	Built-in (via Kortex)	External (optional)
Tool Changer	Compatible	Not standard	Compatible

The xArm 7 holds a clear advantage in end-effector versatility, supporting both parallel jaw and vacuum grippers with factory-compatible accessories. The Gen3 Lite's low payload limits practical gripper options, and the absence of vacuum gripper support is a notable gap for flat-panel e-waste handling.

6. Summary & Recommendation

Based on the technical assessment across specifications, kinematic architecture, end-effector capability, and software integration, the following platform roles are recommended for the E-Waste Disassembly Robot project:

The xArm 7 is the recommended primary platform for physical e-waste disassembly tasks, offering the best combination of payload, speed, DoF, and end-effector versatility. The Kinova Gen3 Lite is best utilised for lighter manipulation experiments and exploratory research. The UR3 serves as the reference simulation platform, enabling rapid prototyping in MATLAB and Isaac Sim prior to physical deployment.

7. Cost Comparison

Component	***xArm 7***	***Kinova Gen3 Lite***	***UR3***
Base Robot	AUD 15,403	AUD 9,500	AUD 18,000
Parallel Gripper	AUD 803	AUD 1,200	AUD 1,500
Vacuum Gripper	AUD 350	N/A	AUD 800
Force/Torque Sensor	AUD 4,444	Built-in	AUD 3,500
ROS/Software	Free	Free	Free
Maintenance (est./yr)	AUD 800	AUD 600	AUD 1,200
Total (approx.)	**AUD 21,800**	**AUD 11,300**	**AUD 24,000**